正文测试：数据配比方法参见 [3]，评测榜单参见 [10]。

参考文献

[1] Qian Liu，Xiaosen Zheng，Niklas Muennighoff，Guangtao Zeng，Longxu Dou，Tianyu Pang，Jing Jiang，Min Lin，RegMix: Data Mixture as Regression for Language Model Pre-training，https://arxiv.org/abs/2407.01492，2026-09-26。

[2] Sang Michael Xie，Shibani Santurkar，Tengyu Ma，Percy Liang，Data Selection for Language Models via Importance Resampling，https://arxiv.org/abs/2302.03169，2026-09-26。

[3] Sang Michael Xie，Hieu Pham，Xuanyi Dong，Nan Du，Hanxiao Liu，Yifeng Lu，Percy Liang，Quoc V. Le，Tengyu Ma，Adams Wei Yu，DoReMi: Optimizing Data Mixtures Speeds Up Language Model Pretraining，https://arxiv.org/abs/2305.10429，2026-09-26。

[4] Sachin Goyal，Pratyush Maini，Zachary C. Lipton，Aditi Raghunathan，J. Zico Kolter，Scaling Laws for Data Filtering – Data Curation cannot be Compute Agnostic，https://arxiv.org/abs/2404.07177，2026-09-26。

[5] Jeffrey Li，Alex Fang，Georgios Smyrnis，Maor Ivgi，Matt Jordan，Samir Gadre，Hritik Bansal，Etash Guha，Sedrick Keh，Kushal Arora，Saurabh Garg，Rui Xin，Niklas Muennighoff，Reinhard Heckel，Jean Mercat，Mayee Chen，Suchin Gururangan，Mitchell Wortsman，Alon Albalak，Yonatan Bitton，Marianna Nezhurina，Amro Abbas，Cheng-Yu Hsieh，Dhruba Ghosh，Josh Gardner，Maciej Kilian，Hanlin Zhang，Rulin Shao，Sarah Pratt，Sunny Sanyal，Gabriel Ilharco，Giannis Daras，Kalyani Marathe，Aaron Gokaslan，Jieyu Zhang，Khyathi Chandu，Thao Nguyen，Igor Vasiljevic，Sham Kakade，Shuran Song，Sujay Sanghavi，Fartash Faghri，Sewoong Oh，Luke Zettlemoyer，Kyle Lo，Alaaeldin El-Nouby，Hadi Pouransari，Alexander Toshev，Stephanie Wang，Dirk Groeneveld，Luca Soldaini，Pang Wei Koh，Jenia Jitsev，Thomas Kollar，Alexandros G. Dimakis，Yair Carmon，Achal Dave，Ludwig Schmidt，Vaishaal Shankar，DataComp-LM: In search of the next generation of training sets for language models，https://arxiv.org/abs/2406.11794，2026-09-26。

[6] Stella Biderman，Hailey Schoelkopf，Quentin Anthony，Herbie Bradley，Kyle O'Brien，Eric Hallahan，Mohammad Aflah Khan，Shivanshu Purohit，USVSN Sai Prashanth，Edward Raff，Aviya Skowron，Lintang Sutawika，Oskar van der Wal，Pythia: A Suite for Analyzing Large Language Models Across Training and Scaling，https://arxiv.org/abs/2304.01373，2026-09-26。

[7] Leo Gao，Stella Biderman，Sid Black，Laurence Golding，Travis Hoppe，Charles Foster，Jason Phang，Horace He，Anish Thite，Noa Nabeshima，Shawn Presser，Connor Leahy，The Pile: An 800GB Dataset of Diverse Text for Language Modeling，https://arxiv.org/abs/2101.00027，2026-09-26。

[8] Guilherme Penedo，Hynek Kydlíček，Loubna Ben allal，Anton Lozhkov，Margaret Mitchell，Colin Raffel，Leandro Von Werra，Thomas Wolf，The FineWeb Datasets: Decanting the Web for the Finest Text Data at Scale，https://arxiv.org/abs/2406.17557，2026-09-26。

[9] Jack W. Rae，Sebastian Borgeaud，Trevor Cai，Katie Millican，Jordan Hoffmann，Francis Song，John Aslanides，Sarah Henderson，Roman Ring，Susannah Young，Eliza Rutherford，Tom Hennigan，Jacob Menick，Albin Cassirer，Richard Powell，George van den Driessche，Lisa Anne Hendricks，Maribeth Rauh，Po-Sen Huang，Amelia Glaese，Johannes Welbl，Sumanth Dathathri，Saffron Huang，Jonathan Uesato，John Mellor，Irina Higgins，Antonia Creswell，Nat McAleese，Amy Wu，Erich Elsen，Siddhant Jayakumar，

Elena Buchatskaya，David Budden，Esme Sutherland，Karen Simonyan，Michela Paganini，Laurent Sifre，Lena Martens，Xiang Lorraine Li，Adhiguna Kuncoro，Aida Nematzadeh，Elena Gribovskaya，Domenic Donato，Angeliki Lazaridou，Arthur Mensch，Jean-Baptiste Lespiau，Maria Tsimpoukelli，Nikolai Grigorev，Doug Fritz，Thibault Sottiaux，Mantas Pajarskas，Toby Pohlen，Zhitao Gong，Daniel Toyama，Cyprien de Masson d'Autume，Yujia Li，Tayfun Terzi，Vladimir Mikulik，Igor Babuschkin，Aidan Clark，Diego de Las Casas，Aurelia Guy，Chris Jones，James Bradbury，Matthew Johnson，Blake Hechtman，Laura Weidinger，Iason Gabriel，William Isaac，Ed Lockhart，Simon Osindero，Laura Rimell，Chris Dyer，Oriol Vinyals，Kareem Ayoub，Jeff Stanway，Lorrayne Bennett，Demis Hassabis，Koray Kavukcuoglu，Geoffrey Irving，Scaling Language Models: Methods, Analysis & Insights from Training Gopher，`https://arxiv.org/abs/2112.11446`，2026-09-26。

[10] Colin Raffel，Noam Shazeer，Adam Roberts，Katherine Lee，Sharan Narang，Michael Matena，Yanqi Zhou，Wei Li，Peter J. Liu，Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer，Journal of Machine Learning Research，21(140)：1–67，2020。

[11] Benjamin Warner，Antoine Chaffin，Benjamin Clavié，Orion Weller，Oskar Hallström，Said Taghadouini，Alexis Gallagher，Raja Biswas，Faisal Ladhak，Tom Aarsen，Nathan Cooper，Griffin Adams，Jeremy Howard，Iacopo Poli，Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference，`https://arxiv.org/abs/2412.13663`，2026-09-26。

[12] Mayee F. Chen，Nicholas Roberts，Kush Bhatia，Jue Wang，Ce Zhang，Frederic Sala，Christopher Ré，Skill-it! A Data-Driven Skills Framework for Understanding and Training Language Models，`https://arxiv.org/abs/2307.14430`，2026-09-26。

[13] Akari Asai，Jacqueline He，Rulin Shao，Weijia Shi，Amanpreet Singh，Joseph Chee Chang，Kyle Lo，Luca Soldaini，Sergey Feldman，Mike D'Arcy，David Wadden，Matt Latzke，Jenna Sparks，Jena D. Hwang，Varsha Kishore，Minyang Tian，Pan Ji，Shengyan Liu，Hao Tong，Bohao Wu，Yanyu Xiong，Luke Zettlemoyer，Graham Neubig，Daniel S. Weld，Doug Downey，Wen-tau Yih，Pang Wei Koh，Hannaneh Hajishirzi，Synthesizing scientific literature with retrieval-augmented language models，Nature，650(8103)：857–863，2026。